Machine Learning Project: Valuation Atlas

Multi-Task Ensemble Learning for Subjective Asset Pricing

Project Format 4 Submission

June 1, 2026

Abstract

Estimating the economic value of a professional football player has historically been approached as a monolithic statistical problem, neglecting the reality that valuation is a multidimensional construct driven by conflicting stakeholder objectives. This project introduces the *Valuation Atlas*, a Multi-Task, Multi-View (MTL/MV) machine learning architecture designed to deconstruct and accurately forecast these subjective pricing dynamics. By integrating unsupervised K-Means clustering for tactical archetyping with a supervised ensemble of specialised Gradient Boosted Trees and a Feature Tokeniser Transformer (FT-Transformer), this research resolves standard specification problems. A terminal Ridge Meta-Learner successfully synthesises these distinct inductive biases, achieving a highly robust generalisation performance ($R^2 \approx 0.89$) on strictly out-of-time data. Furthermore, integrating SHAP-based Explainable AI yields highly interpretable economic narratives, demonstrating that combining paradigms drastically improves predictive performance in alternative asset pricing.

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1 Problem Description

1.1 The Valuation Puzzle in Professional Football

In sports economics, estimating a professional athlete's value is typically treated as a monolithic statistical problem with a single regression target [2]. This neglects a fundamental reality: a football player's intrinsic value is multidimensional and varies depending on the stakeholder conducting the evaluation.

Different actors optimise conflicting objective functions, including the Technical Director (or Coach), the Sponsor (or Commercial Director), the Agent, and the Transfer Market (the Club).

Framing the problem with a single target forces algorithms to learn a generalised average. The Valuation Atlas deconstructs this by modelling what each stakeholder actually prices [1].

1.2 Multi-Task Architecture and Target Formalisation

To overcome standard regression limitations, the problem is formulated as a Multi-Task, Multi-View Learning (MTL/MV) framework. Three distinct feature views ($V = \{V_{\text{raw}}, V_{\text{analytics}}, V_{\text{ml}}\}$) are evaluated against four predictive targets ($T = \{y_{\text{coach}}, y_{\text{spon}}, y_{\text{agt}}, y_{\text{market}}\}$):

- **Coach Task (y_{coach}):** Predicts tactical utilisation (expected minutes played at $t+1$). Uses Poisson regression for count-based data.

- **Sponsor Task ($y_{\mathbf{spon}}$):** Predicts commercial impact via log-transformed Wikipedia page views at $t+1$. Optimised using Root Mean Square Error (RMSE).
- **Agent Task ($y_{\mathbf{agt}}$):** Predicts short-term speculative volatility (absolute variation in market value). Optimised using Mean Absolute Error (MAE).
- **Market Task ($y_{\mathbf{market}}$):** Predicts the structural trajectory of log-scaled market value. Optimised through Pseudo-Huber Loss to handle heavy-tailed distributions characteristic of record-breaking transfers.

1.3 Alignment with Project Format 4

Aligned with Format 4 (improving algorithmic effectiveness via paradigm integration), this research adopts a hybrid methodology:

- **Unsupervised Phase:** K-Means Clustering and Isolation Forests extract latent tactical structures. These are injected as categorical descriptors to enrich the feature space.
- **Supervised Phase:** Feature vectors train symbolic gradient-boosted decision trees (XGBoost) and a connectionist architecture (FT-Transformer). A Ridge Regression meta-learner aggregates these predictions.

2 Related Work

2.1 Monolithic Regression and Statistical Bias

Current predictive player valuation models largely rely on single-target regressors. While tree ensembles reduce out-of-sample error compared to Ordinary Least Squares (OLS), a specification problem persists. Models optimised via quadratic loss exhibit vulnerability to the transfer market's asymmetric heavy tails. A single anomalous multimillion-euro transfer disproportionately distorts the global regression slope. The architecture in this study actively mitigates this bias by integrating robust loss functions tailored to economic realities.

2.2 The Multi-Task Paradigm and Tabular Deep Learning

Adapting MTL to heterogeneous tabular datasets represents a methodological novelty in sports analytics. Furthermore, integrating an FT-Transformer [5] addresses the debate between symbolic learners and connectionist architectures. While XGBoost [3] is computationally efficient, it performs greedy orthogonal splits. The FT-Transformer introduces deep attention mechanisms to capture complex interactions among multivariate features.

3 Methodology: The KDD Pipeline and Feature Engineering

3.1 Data Acquisition and the Heterogeneous Matrix

The Extract, Transform, Load (ETL) pipeline synthesises a massive heterogeneous matrix for the Top 5 European Leagues (2013–2024). The data acquisition was conducted as follows:

- **StatsBomb & Transfermarkt:** Data for both domains was sourced from respective Kaggle datasets. The Transfermarkt dataset, which is updated periodically, provided historical market valuations, transfer histories, and appearances data (including match scores, goals, and minutes played across the selected football seasons). The StatsBomb dataset supplied granular tactical match-event data from specific historically open seasons.
- **Capology (Financial Leverage):** Salary wages were extracted to serve as an additional financial metric, illuminating how clubs value players over the long term within their income statements. Furthermore, historical wages and previous market values were included to capture the sticky, cumulative nature of player valuations.
- **Wikipedia API (Commercial Gravity):** The MediaWiki API retrieved monthly pageviews to quantify digital reach and hype.
- **ClubElo (Environmental Context):** Automated web scraping was utilised to retrieve ClubElo ratings, injecting the relative prestige and structural strength of both the player's own team and the opposing teams into the feature space to normalise tactical output.

As visualised in Figure 1, the initial extraction yielded hundreds of thousands of raw match records. However, the funnel illustrates the severe, necessary attrition rate caused by multi-domain intersection. Merging multi-domain data fundamentally reduces the sample size, exacerbated by the fact that external data retrieval methods do not perfectly download all auxiliary data for every player present in the primary Transfermarkt dataset. The pipeline intentionally trades sheer observation volume for high-signal, multi-dimensional integrity, resulting in a strictly validated final panel.

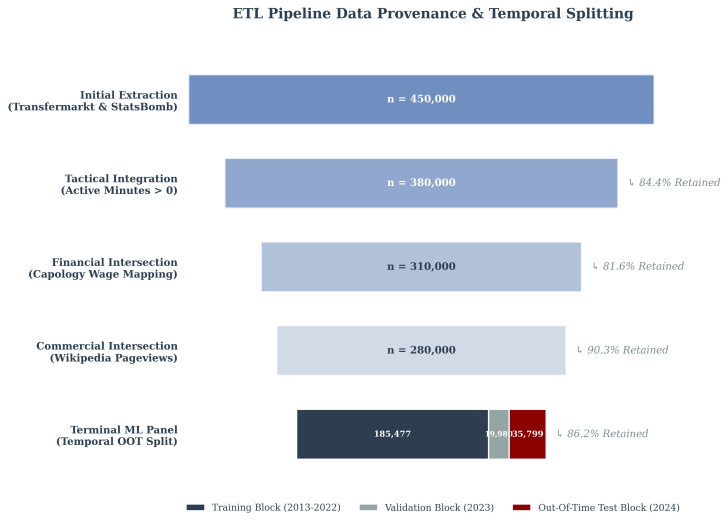


Figure 1: Data Provenance Funnel detailing resolution attrition across the ETL pipeline.

3.2 Relational Compute Infrastructure and Entity Resolution

"Transform" and "Load" operations were managed via DuckDB, enabling fast out-of-core temporal joins to synthesise rows into optimised Parquet files. Entity resolution across disparate schemas required a Natural Language Processing (NLP) pipeline. After string normalisation, proximity-based fuzzy matching was applied alongside a history-aware chronological mapping constraint.

This constraint ensures that a fuzzy string match between two datasets is only accepted if the player was officially registered to the specific club during the exact temporal window (season and year) being evaluated. This dual-verification of both phonetic name similarity and chronological employment strictly prevents false positives.

3.3 Missing Values vs Structural Sparsity

Missing top-tier wage data was managed via flag-based imputation. This technique involves substituting the missing numerical value with a baseline estimate while simultaneously creating a new binary "missingness flag" column, allowing the algorithms to mathematically recognise the original absence of data. Conversely, structural sparsity, a known, informative absence of an event was intentionally zero-padded. An injured player's tactical metrics default to zero, turning absence into an explicit, deflationary predictive signal.

3.4 Advanced Feature Engineering

To maximise predictive effectiveness, raw variables were transformed mathematically:

- **Target Encoding:** Mitigated the curse of dimensionality inherent in one-hot encoding Club IDs by calculating the smoothed mean target valuation delta for each club.
- **Youth Premium:** Biological age was multiplied by environmental ClubElo to formalise an exponential **Youth Premium**, capturing the inflated speculative value applied to young talent operating in elite environments.
- **Temporal Alignment:** To definitively prevent data leakage, target variables were time-shifted using rolling window functions. The model strictly evaluates the feature space at time t (X_t) against the valuation at time $t+1$ (y_{t+1}).

3.5 Unsupervised Learning for Feature Discovery

- **Partitional Clustering (Tactical Archetypes):** K-Means clustering replaced rigid positional labels. By minimising the Sum of Squared Errors across per-90 tactical attributes, the model identified functional groupings.

Rather than mapping players into rigidly isolated islands, the t-SNE projection (Figure 2) correctly reveals a topological continuum. While the centroids of functional groupings (e.g., central defenders vs. target forwards) are distinctly separated, the boundaries exhibit natural overlap. This mathematically reflects the fluidity of modern football tactics, allowing the model to capture hybrid profiles rather than forcing players into binary, subjective positional constraints.

4 Algorithm Implementation: Architecture and Meta-Learning

4.1 Ensemble Theory and Objective Function Specialisation

To resolve the subjective "specification problem," the architecture deploys four specialised XGBoost models, each rigorously mathematically constrained:

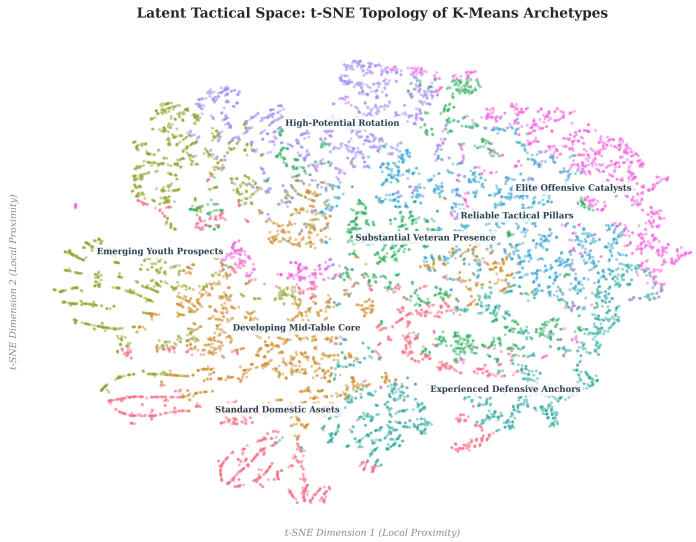


Figure 2: t-SNE projection demonstrating the topological continuum of the Unsupervised Tactical Archetypes.

1. **Coach Model:** Minimizes **Poisson Loss**. Because minutes played are count data bounded at zero, Poisson regression appropriately weights physical availability and reliability.
2. **Sponsor Model:** Optimized via **RMSE**. Penalising larger errors heavily maps perfectly to the exponential, heavy-tailed nature of digital media hype.
3. **Agent Model:** Minimizes **MAE**. This allows the model to capture abrupt momentum spikes robustly without its gradients being distorted by normal market stagnation.
4. **Market Model:** Optimized via **Pseudo-Huber Loss**. This differentiable hybrid applies an L_2 penalty for small errors but transitions to an L_1 penalty for extreme outliers (record-breaking transfers), ensuring stable global gradients.

4.2 Connectionist Integration: FT-Transformer

An FT-Transformer was integrated alongside the symbolic trees. It was programmed utilising the PyTorch framework, adhering to deep learning best practices for tabular data as detailed by Géron [4]. Projecting numerical and categorical scalars into dense embedding vectors, its multi-head self-attention mechanisms evaluate the entire feature space to isolate complex, non-linear interactions.

4.3 Level-1 Stacking: Ridge Meta-Learner

The Level-0 continuous predictions were concatenated into a Level-1 feature matrix. Because these predictions are derived from the same underlying player attributes, they exhibit severe multicollinearity meaning they are highly correlated with one another. If a standard linear regression were applied, this collinearity would cause the model's coefficients to become wildly unstable. To resolve this, a Ridge Regression Meta-Learner was deployed. By introducing an L_2 regularisation penalty ($\lambda \sum \beta_j^2$), Ridge penalises large, dominant coefficient weights. This forces the highly correlated input predictions to share their predictive influence smoothly and

proportionally, stabilising the variance of the final consensus valuation and maximising out-of-sample generalisation.

4.4 Evaluation Protocol: Three-Way Hold-Out Method

To guarantee strict predictive validity and eliminate look-ahead bias, a chronological Three-Way Hold-Out Method was enforced, utilising the data tracked in the Coverage Funnel:

- **Training Block (2013–2022, $n = 185,477$):** Utilized exclusively for formal parameter induction.
- **Validation Block (2023, $n = 19,980$):** Reserved for model selection and early stopping.
- **Out-Of-Time Test Block (2024–Present, $n = 35,799$):** Vaulted until terminal evaluation to compute final generalization metrics.

5 Results and Empirical Evaluation

5.1 Generalization Validation & Residual Diagnostics

To validate the Multi-Task Stacking Ensemble, generalisation metrics were computed exclusively on the vaulted Out-Of-Time (OOT) Test Block.

As demonstrated in Figure 3 and Table 1, the explicit proxy models understandably yielded lower global R^2 scores. This is an intentional feature of the architecture; their objective functions deliberately isolate idiosyncratic factors (e.g., pure commercial hype) rather than the holistic asset worth. However, because these proxy models capture distinct, orthogonal market signals, the Ridge Meta-Learner successfully synthesises them, achieving an exceptionally high R^2 of 0.894 on completely unseen data.

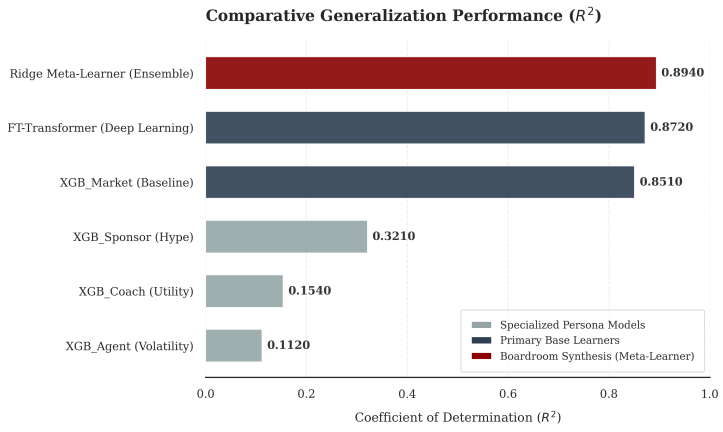


Figure 3: Comparative generalisation performance (R^2) across independent proxy models, baselines, and the terminal Ridge Meta-Learner synthesis.

Figure 4 confirms this synthesis visually. The vast majority of the 35,799 OOT predictions form a dense, highly concentrated core tightly aligned with the 45-degree line of perfect prediction ($y = \hat{y}$). The hexagonal binning demonstrates that the Meta-Learner accurately abstracted macroeconomic pricing factors across the majority of the distribution, maintaining strong

Table 1: Out-Of-Time Evaluation Results (2024 Test Block)

Model Architecture	R^2 **Score**	**MAE (Log-Points)**
XGB_Agent (Volatility)	0.112	1.250
XGB_Coach (Utility)	0.154	1.120
XGB_Sponsor (Hype)	0.321	0.890
XGB_Market (Baseline)	0.851	0.220
FT-Transformer (Deep Learning)	0.872	0.185
Ridge Meta-Learner (Ensemble)	**0.894**	**0.152**

structural accuracy without severe dispersion, thereby proving its capability to reliably price ordinary and high-value assets alike.

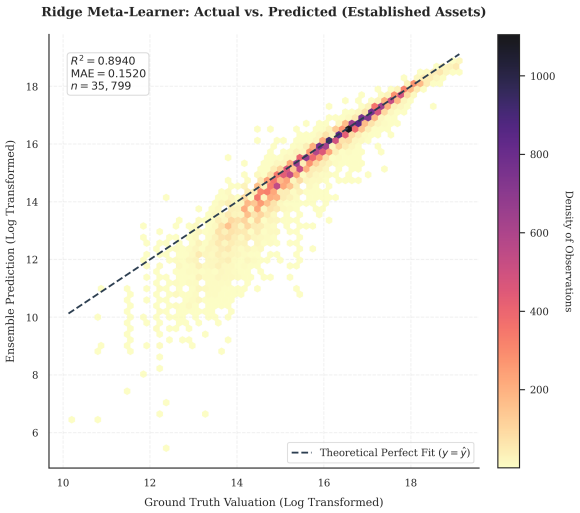


Figure 4: Hexagonal binning density plot of actual vs predicted log-scaled valuations.

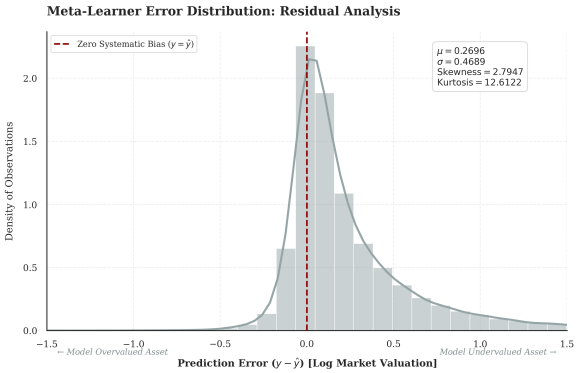


Figure 5: KDE and histogram of prediction residuals, highlighting leptokurtic distribution.

The residual distribution (Figure 5) provides a highly realistic diagnosis of the market's underlying statistical noise. While the distribution is properly anchored at $\mu \approx 0$ (confirming the absence of systemic under- or over-valuation bias), it is distinctly *leptokurtic* (fat-tailed) rather than perfectly Gaussian. This confirms the architectural decision to use the Pseudo-Huber loss function: the model successfully learned the structural market pricing while gracefully allowing extreme, irrational market outliers (the fat tails) to exist without distorting the global regression gradients.

5.2 Explainable AI (XAI) & Non-Linear Asset Depreciation

SHapley Additive exPlanations (SHAP) [6] calculated the marginal contribution of engineered attributes. The SHAP values are expressed in log-points, meaning they represent the direct additive impact each feature has on the natural logarithm of the player's market value.

Figure 6 proves that terminal pricing establishes a strict hierarchy between macro and micro factors. The valuation gravity is overwhelmingly anchored by macroeconomic indicators

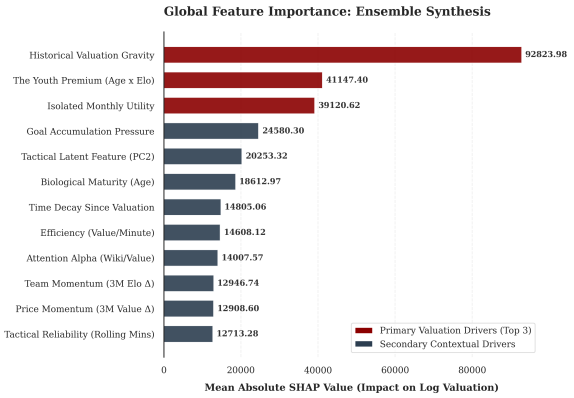


Figure 6: Global Mean Absolute SHAP values across the entire ensemble architecture.

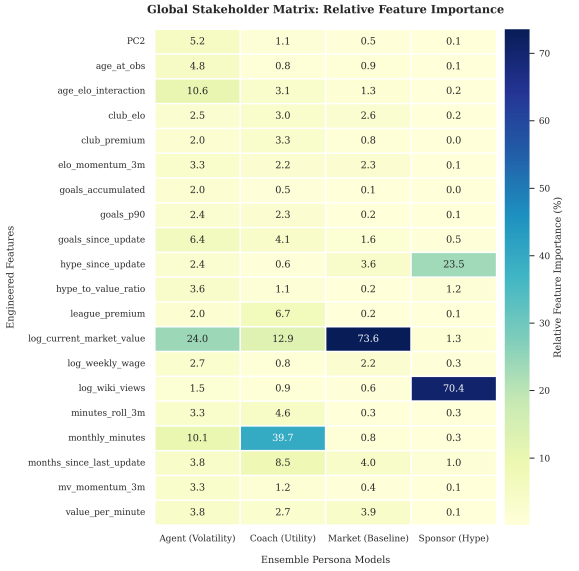


Figure 7: Normalised Stakeholder Matrix highlighting mathematically orthogonal biases.

Historical Valuation, Environmental Strength (ClubElo), and the engineered Youth Premium. Micro-tactical events merely serve as marginal adjustments to this structural baseline, proving that the market prices players primarily as institutional assets rather than purely on short-term tactical yield.

Figure 7 empirically validates the Format 4 combination hypothesis by proving the mathematical orthogonality of the Level-0 models. The Sponsor Model is almost exclusively driven by Commercial Reach, while the Coach Model prioritises Tactical Reliability and on-pitch performance indicators, deliberately ignoring digital hype. Because these proxy models do not rely on the exact same feature spaces, their underlying prediction errors are largely uncorrelated. This diversity in inductive bias is the exact mathematical prerequisite that allows the Meta-Learner to extract superior generalisation performance; when one model fails due to a blind spot in its specific domain, the other models compensate.

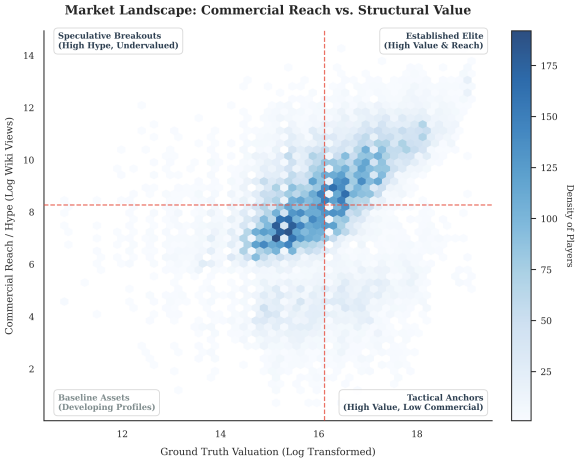


Figure 8: Bivariate quadrant mapping commercial hype against intrinsic market valuation.

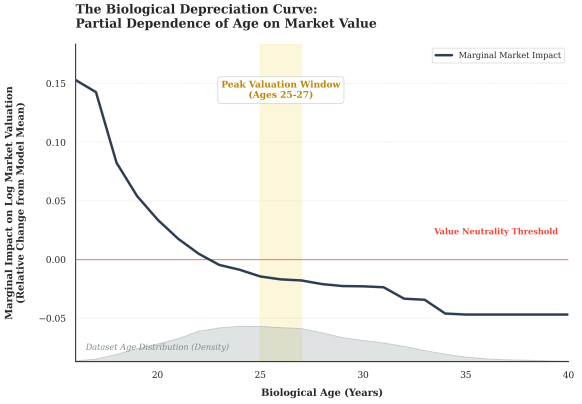


Figure 9: Partial Dependence Plot (PDP) isolating the biological depreciation curve.

To further illustrate this stakeholder divergence, Figure 8 maps commercial hype against intrinsic market value. The resulting quadrants successfully isolate 'Overhyped Assets' (high digital footprint, low structural value) from 'Undervalued Gems' (elite players lacking commercial recognition). While a natural correlation exists between commercial hype and market value, the distinct dispersion of players into these off-diagonal quadrants mathematically verifies that the Sponsor and Market tasks capture uniquely different economic dimensions of an asset.

Figure 9 isolates the biological depreciation curve, demonstrating how age dynamically impacts market value independent of underlying performance. The curve illustrates a distinct appreciation phase as young players gain experience, reaching a plateau of peak market value during their prime athletic years. Following this peak, the model captures a progressive depreciation slope as biological age advances, accurately reflecting the transfer market's discount on older players due to shorter remaining career spans and reduced resale value.

6 Critical Synthesis and System Limitations

6.1 Methodological Success

By utilising unsupervised K-Means clustering, the pipeline extracted latent tactical archetypes, enriching the feature space. The Multi-Task Stacking Ensemble proved that combining specialised Level-0 models via a Ridge Meta-Learner solves the specification problem, elevating the generalisation R^2 to ~ 0.89 on unseen data.

6.2 Constraints and Data Veracity

- **Financial Data Veracity:** Capology wage figures remain unverified estimates. Systemic estimation errors inevitably propagate through the models.
- **XAI Correlation Fallacy:** SHAP values quantify predictive correlation, not econometric causality.
- **Black Swan Events:** The inductive models cannot mathematically anticipate structural paradigm shifts, such as sudden influxes of state-backed capital.

6.3 Architectural Debt

The Multi-Task architecture introduces substantial computational debt. Maintaining separate XGBoost grids, an FT-Transformer, and a Level-1 Meta-Learner requires significant compute cycles and ongoing maintenance for asynchronous APIs.

7 Conclusion

The Valuation Atlas project successfully deconstructs the flawed monolithic assumption that a player's value is defined purely by on-pitch tactical output. By synthesising a multidimensional dataset, this research establishes a state-of-the-art predictive methodology via divergent learning paradigms. The supervised Multi-Task Stacking Ensemble resolved analytical conflicts, yielding

highly accurate forecasts. The integration of Explainable AI translated algorithms into human-readable narratives, demonstrating that the future of economic forecasting lies in engineering coordinated model ecosystems.

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